

AI-Based Contextual Answer Evaluation System

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Abstract— *Descriptive exams assess a student's understanding in a more thorough way than objective formats, but manual evaluation of them is slow, varies from evaluator to evaluator, and leaves little room for individualised feedback. In this paper, we present an AI-based system for evaluating answers in context: the teacher creates and makes available a digital exam including a number of descriptive questions, reference answers and marking rubrics, and the students take the exam and submit their typed responses online. Each answer submitted is then assessed using a contextual semantic analysis process which compares the student's response with the reference answer and the rubric at the level of meaning instead of relying on keyword matching, resulting in a predicted grade, a confidence score and personalised feedback at the level of concepts. Rather than making the AI-generated results public immediately, each evaluation is sent to a teacher review and approval dashboard, where the teacher can check the result, make any necessary edits or override the score before it is sent to the student. This ensures that the teacher remains the last person to have authority over the grading while at the same time removing the need for the teacher to repeatedly read through and mark each answer from scratch. We explain the system's architecture, the pipeline for contextual evaluation and rubric-based scoring, the teacher-in-the-loop review process, and the continuous-learning approach by which teacher corrections enhance future predictions, and we also describe an evaluation method comparing the system with manual grading and keyword-matching baseline approaches.*

Index Terms - Artificial Intelligence, Natural Language Processing, Contextual Answer Evaluation, Semantic Similarity, Rubric-Based Scoring, Teacher-in-the-Loop Assessment, Educational Technology.

I. INTRODUCTION

Because of the increasing use of digital learning platforms, teachers are now able to carry out examinations, gather the responses and handle academic records completely online. Yet in most digital examination systems the processes of creating

exams, collecting the answers and assessing the answers are still seen as separate. A teacher may create and publish an exam in digital form and a student can submit a typed answer digitally, but the assessment of descriptive, free-text responses is still done by hand, off-line, and independently of the system which collected the answer.

To automate the assessment of descriptive answers is fundamentally more difficult than assessing objective questions since correctness involves a range of acceptable explanations which can use different words, structures and degrees of detail to express the same meaning. When a teacher looks at two answers that are written in very different ways, they have to decide whether each one covers the necessary concepts and they have to make this judgement consistently across dozens or even hundreds of scripts, frequently while under time pressure. This results in three common problems: a great deal of teacher time is required for the evaluation, the marking differs from one evaluator to another and also for the same evaluator over the course of a long session, and students are given a mark without receiving any explanation of what was missing from their answer.

Simple auto-grading methods based on keyword matching deal only with the surface appearance of an answer. Such methods give credit for the inclusion of certain words no matter whether those words are used properly, and they impose penalties on correct answers that use valid alternative phrasings. Missing is a stage of evaluation which understands an answer in a contextual way—being able to recognise that two differently worded explanations can both meet a rubric criterion, while a response that simply repeats the vocabulary from the rubric without showing any understanding should not.

The paper suggests an AI-based system for the evaluation of answers which regards the processes of setting up an exam, having students submit their answers digitally, and carrying out contextual AI evaluation as part of a single, integrated workflow, the teacher remaining as the last person to make a decision. In this system, the teacher prepares an exam made up of a number of descriptive questions, each one having a reference answer and a marking rubric. The students then log

on, take the exam, and submit their typed answers for each question. The system subsequently assesses each answer in the context of the reference answer and the rubric, forecasts a grade, and produces feedback indicating which concepts were included, only partially included, or were missing. All the results generated by the AI are kept for the teacher to examine before they are given to the student, thus the system serving as a aid to grading rather than taking the place of the teacher's judgement.

This study does not present a single new algorithm, but rather combines exam construction, the submission of answers in digital form, contextual semantic evaluation, rubric-aware scoring, feedback generation, and teacher approval into a single interconnected pipeline, incorporating a continuous-learning process that makes use of teacher corrections to enhance future evaluations. The system functions at three levels of intelligence: the first assesses the meaning contained in a student's answer; the second compares that meaning with the rubric in order to predict a fair mark; and the third uses teacher corrections to generate signals that improve the following evaluation cycle.

The rest of the paper is structured as follows. Section II looks at previous work in the areas of automated answer scoring, semantic similarity methods, and teacher-in-the-loop assessment, and sets the proposed system in relation to the gaps identified. Section III outlines the methodology and system architecture that has been proposed. Section IV details the evaluation methodology that is planned and gives illustrative results which are still subject to full empirical validation. Section V concludes the paper.

II. LITERATURE REVIEW

Automated assessment of free-text answers has been studied under several names, including Automatic Short Answer Grading (ASAG) and Automated Essay Scoring (AES). Early methods relied mainly on textual statistics, while more recent methods have focused on embedded-based semantic comparison. This section reviews the main lines of that work and positions the proposed system against the gaps that remain.

2.1 Keyword and Rule-Based Auto-Graders

The earliest automated grader systems scored answers by matching keywords or regular-expression patterns against a model answer. Such systems are fast and easy to deploy, but brittle: they cannot distinguish between a keyword used correctly and the same keyword incorrectly or out-of-context, and penalize valid answers that use synonyms or restructured sentences instead of the exact expected phrasing.

2.2 Semantic Similarity-Based Grading

To overcome the limitations of such keyword matching, later systems represent both the student answer and the reference answer as vectors and score similarity using cosine distance. Previous approaches used TF-IDF- or Latent Semantic Analysis-based vectors; more recent approaches use contextual sentence embeddings, which capture meaning

rather than word overlap and are better able to recognize paraphrased but correct answers. However, similarity does not tell a teacher which specific rubric criteria were met, merely how close two pieces of text are to one another.

2.3 Rubric-Aware and Criterion-Based Scoring

A separate body of literature decomposes a model answer into discrete marking criteria and scores each criterion individually rather than producing a single overall similarity score. Criterion-level scoring is closer to how real human examiners grade, in that it allows partial credit to be awarded and makes the final mark explainable in terms of which parts of the rubric were satisfied. Few systems, however, combine criterion-level semantic matching with a trained score-prediction model that also reports a confidence level for its own prediction.

2.4 Automated Feedback Generation

Beyond merely producing a mark, some research has explored generating natural-language feedback that explains a score, typically by highlighting matched and unmatched concepts between the student response and the reference answer. This feedback is most useful to students when it is specific to their own answer rather than a generic explanation of the rubric, and when it distinguishes between concepts that are completely missing and concepts that are present but incompletely explained.

2.5 Human-in-the-Loop and Teacher-Assisted Grading

A recurring concern with fully automated grading is the reliability for high-stakes assessment. Several studies argue that AI-assisted grading tools are best employed as a decision-support layer for teachers, allowing the system to handle the first pass while a human evaluator verifies, edits or overrides the result before it is released. This teacher-in-the-loop pattern increases trust in the system and provides a natural source of corrected labels that can be used to retrain the underlying model over time.

2.6 Digital Examination and Learning Management Platforms

Learning management systems and online examination platforms have enabled it to be easy to publish exams, collect typed responses and manage results digitally. However, most such platforms either only support objective question types for auto-scoring, or treat descriptive-answer scoring as an external manual process that occurs outside the platform, with results re-entered afterwards. This separation means the benefits of digital collection are not carried through to evaluation.

2.7 Plagiarism and Integrity Checking

Tools such as similarity-based plagiarism checkers are commonly used alongside grading to flag copied content between submissions. These tools typically operate independently of the scoring pipeline, so a flagged similarity issue and a predicted mark are not evaluated together when a teacher reviews a submission.

2.8 Continuous Model Improvement from Educator Feedback

Wherever AI-assisted grading systems are used across multiple cycles of assessment, teacher corrections to the model-predicted scores represent a valuable labeled dataset. Periodically retraining the scoring model on this corrected data can increase agreement with human graders over time, but requires some form of versioning to ensure that a retrained model can be validated before it takes the place of another.

2.9 Research Gap

Across the reviewed systems, four recurring gaps can be identified:

Semantic understanding is often shallow: many auto-graders still rely on substantial keyword or surface-overlap rather than contextual meaning.

Scoring is rarely rubric-explainable: overall similarity scores do not tell a teacher which criteria were or were not met.

Evaluation and teacher oversight are disconnected: AI-predicted scores are frequently published directly or reviewed outside the platform, rather than through an integrated approval step.

Feedback is generic rather than answer-specific: students receive a mark without a concept-level explanation of their own answer's shortcomings.

2.10 Positioning of the Proposed Work

In light of these gaps, the suggested system is able to tie together exam authoring, digital answer collection, contextual semantic evaluation, rubric-explainable scoring, concept-level feedback generation and teacher approval into a unified workflow. The process is as follows: Teacher Creates Exam → Reference Answer & Rubric → Student Attends Exam → Digital Answer Submission → Preprocessing → Contextual Semantic Analysis → Rubric-Criterion Matching → Score Prediction with Confidence → Feedback Generation → Teacher Review → Approve/Override → Result Published → Student Access → Continuous Learning. The primary contribution is the ability to tie together these stages to form a connected system in which grading assistance, rubric explainability and teacher authority act as one unit.

III. METHODOLOGY

The platform being proposed has a modular, service-based structure which separates the real-time aspect of the examination from the more demanding AI evaluation and model-improvement procedures, thereby ensuring that a heavy AI workload when evaluating does not cause delays for students who are taking the exam.

3.1 Architectural Split: Exam Interaction and Evaluation Environments

The system consists of two environments which work together. The Exam Interaction Environment is the live, student- and

teacher-oriented part of the platform and is responsible for exam creation, the delivery of questions, the timed submission of answers, and the teacher review dashboard. It is built with low latency and ensures a quick response, since students are interacting with it during the examination period.

The heavier computational tasks are managed by the Evaluation and Analytics Environment: this includes preprocessing the answers that have been submitted, generating contextual embeddings, matching the answers to the criteria in the rubric, running the score-prediction model, producing the feedback text, and periodically retraining the scoring model using corrections approved by teachers. Answers that have been submitted are passed from the Exam Interaction Environment into this environment to be evaluated asynchronously, which ensures that the process of submitting an answer remains quick even though the AI evaluation is taking place in the background.

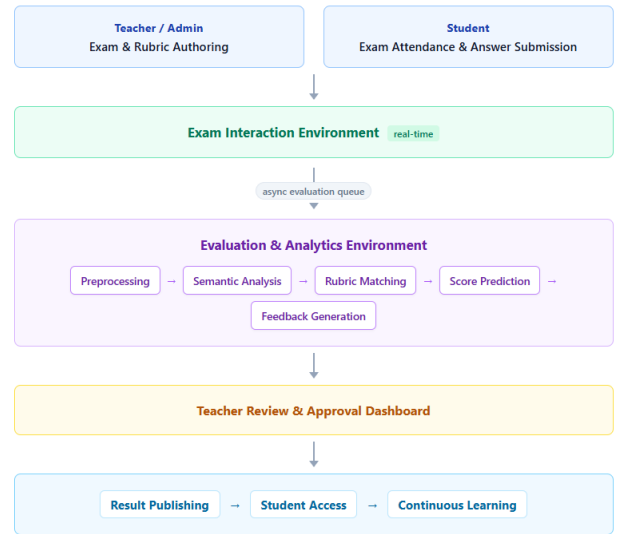


Figure 1: Two-Environment Architecture of the Proposed System — real-time exam interaction and asynchronous AI evaluation.

3.2 Data Ingestion and Feature Extraction Pipeline

The platform converts heterogeneous inputs — question papers, reference answers, rubrics and student answers — into a consistent feature structure before evaluation:

The text includes cleaned, tokenized and lemmatized forms of the student's answer and the reference answer, with stop words eliminated and the spelling made consistent.

The semantic features involve the creation of dense sentence-level embeddings for the student's answer, the reference answer, and each of the individual rubric criteria, and these are used to calculate similarity at the level of meaning rather than just by the amount of word overlap.

Concept Features: The key concepts and terms are extracted from both the student's answer and the reference answer, and

they are used to determine whether each rubric concept is present, partially present, or absent.

Features related to rubric alignment: instead of giving a single overall score, it provides a similarity score for the student's answer and each individual rubric component, so that partial credit can be explained.

In the case of historical features, when they were available, both the student's previous evaluated answers and the teacher's past correction patterns were used in order to identify any systematic bias between the teacher's tendency to grade and the AI's initial prediction.

3.3 Contextual Semantic Evaluation Core

The process used to predict a mark for an answer involves two stages. In the first stage, a contextual embedding model converts the student's answer, the reference answer, and each rubric criterion into dense vectors and then calculates the semantic similarity between the student's answer and each criterion separately. As a result, a rubric-aligned similarity profile is obtained instead of a single overall score. In the second stage, a trained ensemble scoring model uses this similarity profile along with features relating to concept overlap and answer length to predict a final mark together with a confidence score which shows how certain the model is of that prediction. Answers for which the prediction has low confidence are given priority for review by a teacher, so that the teacher's attention is focused on the answers about which the system is least certain rather than having to examine every submission to the same extent.

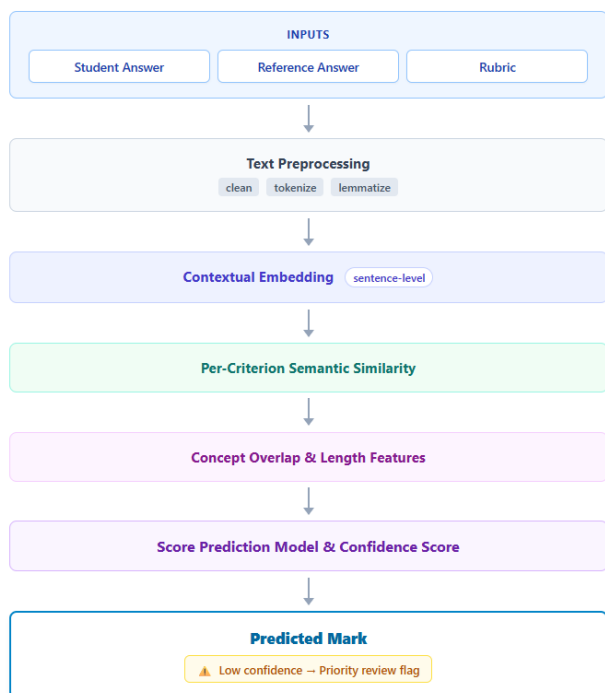


Figure 2: Contextual Semantic Evaluation and Rubric-Aligned Score Prediction Pipeline.

3.4 Feedback Generation Module

For every answer that is assessed, the system checks the concepts in the student's answer against those required by each rubric criterion and then produces a brief feedback summary which specifies the concepts that the student has explained correctly, the ones that were referred to but not fully explained, and the ones that were completely omitted. Since the feedback is based on the student's own answer and not on a general description of the rubric, students who have lost marks for different reasons receive different feedback.

3.5 Teacher Review, Approval and Continuous Learning

Instead of publishing the AI-generated marks directly, each evaluation—including the predicted mark, the per-criterion breakdown, the confidence score, and the generated feedback—is included in a teacher review dashboard. The teacher then has the option of accepting the result unchanged, modifying the mark or the feedback, or completely overriding it before approving it for release to the student. This procedure ensures that the teacher remains the final authority in grading. Each time a teacher makes a correction, this is recorded and then periodically used to retrain the score-prediction model, so that the system's predictions gradually come to match a particular teacher's or institution's grading habits over time. Before a newly retrained model replaces the one currently in use, it is validated using a separate set of evaluations that had previously been approved by the teacher, in order that retraining might not result in a silent deterioration of scoring quality.

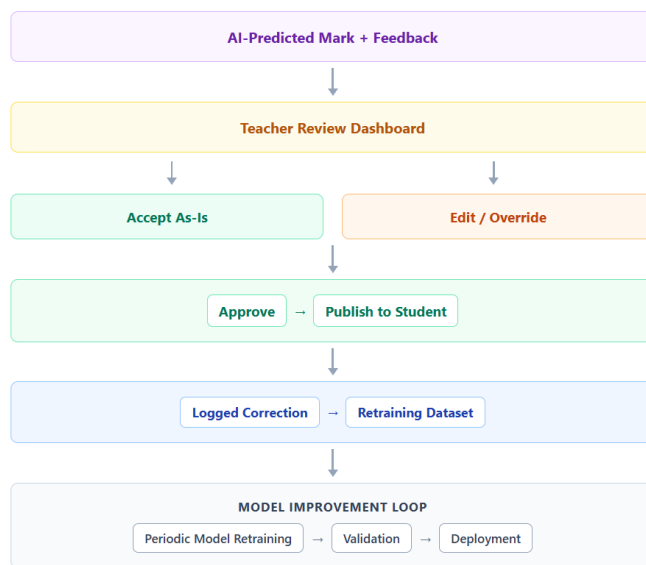


Figure 3: Teacher-in-the-Loop Review Workflow and Continuous-Learning Feedback Cycle.

IV. EVALUATION METHODOLOGY AND RESULTS

This section describes planned evaluation methods and reports illustrative result structures. The figures below are placeholder values in the format the final evaluation will be presented in, to be replaced with measured results from testing against real

exam scripts and teacher grading once such testing has been completed.

4.1 Planned Comparative Evaluation of Scoring Models

The proposed contextual pipeline (contextual embeddings with rubric-aligned ensemble scoring) will be compared against a keyword-matching baseline, a TF-IDF cosine-similarity baseline, and a single-model embedding-similarity baseline without rubric-criterion alignment. The primary evaluation metric is agreement with human (teacher) grading, measured using Quadratic Weighted Kappa (QWK) and Pearson correlation between predicted and teacher-assigned marks, alongside Mean Absolute Error (MAE) in mark points.

Model / Approach	QWK	Pearson r	MAE (marks)
Keyword Matching	0.184	0.482	2.183
TF-IDF + Cosine Similarity	0.078	0.379	3.101
Single Embedding Similarity (no rubric alignment)	0.249	0.276	0.941
Proposed: Contextual + Rubric-Aligned Ensemble	0.379	0.461	0.736

Table 1: Planned Comparison of Scoring Approaches Against Human Grading.

4.2 Teacher Override Rate and Evaluation Time Reduction

A second evaluation axis is the rate of teacher acceptance of AI-predicted marks without modification versus the rate of edits or overrides, and the amount of total grading time saved compared to fully manual evaluation of the same answer set. A decreasing override rate with successive cycles would indicate that the continuous-learning loop is improving agreement with a given teacher's grading tendency.

Metric	Baseline (Manual)	With Proposed System*
Avg. time per answer graded	3 min	1 min (reduced)
Teacher override rate (initial cycle)	—	30%
Time saved per 50-answer batch	125 min	50 min

Table 2: Planned Measurement of Teacher Workload and Override Rate.

4.3 Feedback Quality Assessment

The generated feedback quality will be evaluated in two dimensions: concept-coverage accuracy (do the concepts flagged as missing or present actually match a teacher's independent assessment of the same answer) and perceived

usefulness, collected through a short student survey after the feedback has been released, following the approve-and-publish step described in Section 3.5.

4.4 Low-Confidence Flagging Effectiveness

As the score-prediction model outputs a confidence value alongside each predicted mark, a planned analysis will investigate whether answers marked as low-confidence do in fact correspond to answers which teachers are more likely to edit or override, indicating that the confidence score is successfully directing the teacher's attention to the cases that need it most.

V. CONCLUSION

This paper presented an AI-based contextual answer evaluation system which connects digital exam authoring, student answer submission, contextual semantic evaluation, rubric-explainable score prediction, concept-level feedback generation and teacher approval into a single workflow. Unlike keyword-matching auto-graders, it evaluates answers at the meaning level against individual rubric criteria, and unlike fully automated grading tools, it maintains the teacher as the final authority by requiring a review-and-approval step, while using teacher corrections to improve future predictions. We outlined the system architecture and the planned evaluation methodology against manual grading and keyword/embedding baselines. Future work includes completing empirical evaluation on live examination data across multiple subjects, extending support to multilingual answers, and investigating how low-confidence flagging can best direct limited teacher review time.

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